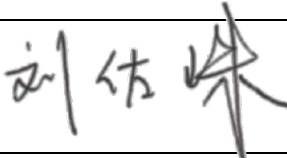


Appendix V

Midterm Self-check Form

Student Name:	Zihan Xu	ZJU ID	3220110781	Major	CompE
Supervisor	Zuozhu Liu		Academic Title	Assistant Professor	
Project Title	Medical Reasoning with Memory-Augmented Agentic System				
Tasks Completed	Description (Statement of thesis progress, No more/less than 500 word) At the current stage, we have completed the design and constructed an evaluation framework for long-horizon clinical decision-making from real-world Electronic Health Record (EHR) data (<i>MIMIC-IV</i>). First, at the data level, we designed a comprehensive data processing pipeline. By filtering for patients with multiple hospitalization records, we curated a clinical trajectory dataset characterized by long time spans and multi-session properties, thereby establishing a foundation for evaluating long-horizon reasoning capabilities. Building upon this foundation, we proposed a three-tiered memory structure (<i>Note Memory, Event Memory, and Contextual Memory</i>), each corresponding to historical information at a distinct granularity. The memory supports the agent performing information retrieval and reasoning across varying contextual scopes. In terms of task design, we formulated <i>Long-horizon Decision Making</i> tasks to simulate the selection of next actions in realistic clinical scenarios. We provided the agent with various methods of memory organization at different granularities to comprehensively assess the capability boundaries of medical agents. In the experiments, we conducted preliminary evaluations utilizing various Large Language Models (LLMs) and also compared the performance of different memory augmentation methods (e.g., RAG and Agent Memory Systems). The experimental results indicate that memory augmentation methods yield only limited improvements in decision-making tasks, and the performance of agent decision-making depends more on the immediate interaction context. Overall, we have successfully completed the data construction, task design, and preliminary experimental validation phases, thereby laying foundation for subsequent in-depth research.				
	Compliance with Task Schedule as per Project Plan (To be filled by the Supervisor (Y/N))				Y

Task(s) to be completed	Explore designs for long-horizon decision-making tasks of higher quality and refine problem generation to enhance data reliability and evaluation stability. Categorize task types among existing research within EHR domain, distinguishing between multi-turn and multi-session task settings. Refine existing task taxonomies and delineate the innovative aspects from existing work and the novel tasks introduced in this project. Establish standardized data processing and experimental workflows—including the standardization of data extraction, preprocessing, task generation, and evaluation scripts—to realize an end-to-end, reproducible pipeline.		
Problems and Solutions	Problem 1: EHR data is characterized by complex structures and significant noise, which may lead to inconsistencies in task construction. Solution: We designed a multi-stage data cleaning and filtering pipeline to retain only critical information—such as abnormal test results and complete hospitalization records—and standardized the format for representing clinical events, thereby enhancing data quality. Problem 2: some benchmark questions exhibiting unreasonable difficulty levels or semantic flaws. Solution: We introduced a dual-filtering mechanism—leveraging both rule-based methods and Large Language Models (LLMs)—to validate the semantics and regulate the difficulty of generated questions, while simultaneously designing automated evaluation metrics to facilitate the selection process. Problem 3: the definition of long-range tasks lacks clarity, and the boundaries between "multi-turn" and "multi-session" interactions remain ambiguous. Solution: Through a systematic review of relevant literature, we developed a hierarchical modeling approach for these tasks. This approach clearly delineates the input-output structures and evaluation objectives for different task types, culminating in the establishment of a unified framework for task classification.		
Supervisor signature		Date	Mar. 24, 2026